After class: check your understanding

Dr. NAM · Faculty of Pharmacy, UiTM Bertam, Penang

About 30 minutes. Questions follow the subchapters of the lecture and end with a pharmacy scenario.

Answer without notes first, then use the slides to check. The answer key is at the back.

8.1–8.2 · Principles

Q1 CH8-LO1

Match each type of chromatography to how the solute is held.

Options: sticks to a solid surface · dissolves in a liquid film · attracted to fixed charges · large molecules come out first

adsorption _____

partition _____

ion exchange _____

size exclusion _____

Q2 CH8-LO1

'Gas chromatography' tells you that the:

A. mobile phase is a gas

B. stationary phase is a gas

C. sample is a gas only

D. detector uses gas

Q3 CH8-LO1

A peak has t_r = 5.0 min and t_m = 1.0 min. What is its retention factor k?

Q4 CH8-LO1

Two peaks are fully (baseline) separated when the resolution R is at least:

A. 1.5

B. 0.5

C. 1.0

D. 10

8.3 · Stationary and mobile phases

Q5 CH8-LO2

Match each stationary phase to its polarity or use.

Options: polar adsorbent · nonpolar, reversed phase · holds water, partition · cation exchanger

silica gel

C18 bonded silica

cellulose paper

sulfonated resin

Q6 CH8-LO2

In reversed-phase HPLC, which elutes first?

A. the more polar compound

B. the less polar compound

C. the larger molecule

D. the compound with the highest boiling point

Q7 CH8-LO2

Calculate P′ for a 50:50 water–methanol mobile phase (water 10.2, methanol 5.1). Give one decimal place.

Q8 CH8-LO2

In gradient elution, the composition of the mobile phase changes during the run.

True / False

8.4 · Thin layer chromatography

Q9 CH8-LO3

A spot moves 2.4 cm and the solvent front moves 8.0 cm. What is R_f?

Q10 CH8-LO3

Why must the solvent level in the tank be below the origin line?

A. so the spots are not washed off into the solvent pool

B. so the plate dries faster

C. so the R_f is higher

D. to save solvent

Q11 CH8-LO3

Match each way of showing spots to what it reveals.

Options: compounds that absorb UV (dark spots) · amino acids (purple) · phenols (coloured) · alkaloids (orange)

UV light, 254 nm

ninhydrin spray

iron(III) chloride spray

Dragendorff's reagent

Q12 CH8-LO3

A co-spot of a test sample with the standard gives one single spot. This:

A. supports that the test drug is the same as the standard

B. proves the sample is 100% pure

C. means the solvent was wrong

D. means the sample contains no drug

8.5–8.7 · Ion exchange, HPLC and GC

Q13 CH8-LO4

A weak acid cation exchanger carries which group?

A. $-COO^-$

B. $-SO_3^-$

C. $-N(CH_3)_3^+$

D. $-NH_3^+$

Q14 CH8-LO4

Match each part of an HPLC to its job.

Options: pushes the mobile phase at high pressure · delivers an exact volume of sample · separates the components · signals each compound as it leaves

pump _____

loop injector _____

C18 column _____

UV detector _____

Q15 CH8-LO4

Match each GC detector to what it is best for.

Options: almost all organic compounds · halogenated compounds · everything, but less sensitive · identifying unknowns

FID _____

ECD _____

TCD _____

MS _____

Q16 CH8-LO4

Temperature programming in GC means:

A. the oven heats up during the run

B. the injector is cooled

C. the carrier gas is changed

D. the sample is frozen first

Pharmacy scenario · The slimming capsule

Scenario

Puan Rosnah brings a 'natural slimming' capsule bought online and says it makes her heart race. The pharmacist reports it to the regulator, whose lab suspects a hidden drug. The lab also tests a returned bottle of **paracetamol tablets** that were stored in a hot car.

Q17 CH8-LO5

The quickest, cheapest first screen for a hidden drug in the capsule is:

A. TLC against reference standards of suspected drugs

B. GC of the capsule shell

C. ion-exchange chromatography

D. size-exclusion chromatography

Q18 CH8-LO5

To confirm the identity of the hidden drug beyond doubt, the lab would use:

A. LC–MS or GC–MS
B. paper chromatography
C. a melting point only
D. iodine vapour

Q19 CH8-LO5

For the paracetamol tablets, which test shows whether the drug has broken down?

A. HPLC: less paracetamol and new breakdown peaks
B. GC residual solvents
C. a co-spot only
D. ion chromatography of chloride

Q20 CH8-LO5

Explain to Puan Rosnah, in plain words, how the lab can tell what is inside her capsule.

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About 30 minutes. Questions follow the subchapters of the lecture and end with a pharmacy scenario.

Answer key

8.1–8.2 · Principles

Q1 CH8-LO1

Match each type of chromatography to how the solute is held.

adsorption → sticks to a solid surface; partition → dissolves in a liquid film; ion exchange → attracted to fixed charges; size exclusion → large molecules come out first

Harvey 12.1.

Q2 CH8-LO1

'Gas chromatography' tells you that the:

A. mobile phase is a gas

B. stationary phase is a gas

C. sample is a gas only

D. detector uses gas

If only one phase is named, it is the mobile phase.

Q3 CH8-LO1

A peak has t_r = 5.0 min and t_m = 1.0 min. What is its retention factor k?

4

k = (5.0 − 1.0) ÷ 1.0 = 4.0.

Q4 CH8-LO1

Two peaks are fully (baseline) separated when the resolution R is at least:

A. 1.5

B. 0.5

C. 1.0

D. 10

Harvey 12.2.

8.3 · Stationary and mobile phases

Q5 CH8-LO2

Match each stationary phase to its polarity or use.

silica gel → polar adsorbent; C18 bonded silica → nonpolar, reversed phase; cellulose paper → holds water, partition; sulfonated resin → cation exchanger

8.3 and 8.5.

Q6 CH8-LO2

In reversed-phase HPLC, which elutes first?

A. the more polar compound

B. the less polar compound

C. the larger molecule

D. the compound with the highest boiling point

The nonpolar C18 holds less polar compounds longer.

Q7 CH8-LO2

Calculate P′ for a 50:50 water–methanol mobile phase (water 10.2, methanol 5.1). Give one decimal place.

7.7

0.5(10.2) + 0.5(5.1) = 7.65 ≈ 7.7.

Q8 CH8-LO2

In gradient elution, the composition of the mobile phase changes during the run.

True / False → **True**

Isocratic elution keeps it the same.

8.4 · Thin layer chromatography

Q9 CH8-LO3

A spot moves 2.4 cm and the solvent front moves 8.0 cm. What is R_f?

0.3

2.4 ÷ 8.0 = 0.30.

Q10 CH8-LO3

Why must the solvent level in the tank be below the origin line?

A. so the spots are not washed off into the solvent pool

B. so the plate dries faster

C. so the R_f is higher

D. to save solvent

The spots would dissolve into the solvent instead of travelling up.

Q11 CH8-LO3

Match each way of showing spots to what it reveals.

UV light, 254 nm → compounds that absorb UV (dark spots); ninhydrin spray → amino acids (purple); iron(III) chloride spray → phenols (coloured); Dragendorff's reagent → alkaloids (orange)

8.4.

Q12 CH8-LO3

A co-spot of a test sample with the standard gives one single spot. This:

A. supports that the test drug is the same as the standard

B. proves the sample is 100% pure

C. means the solvent was wrong

D. means the sample contains no drug

Different compounds would usually split into two spots.

8.5–8.7 · Ion exchange, HPLC and GC

Q13 CH8-LO4

A weak acid cation exchanger carries which group?

A. $-COO^-$

B. $-SO_3^-$

C. $-N(CH_3)_3^+$

D. $-NH_3^+$

Carboxylic acid: weak; sulfonic acid: strong.

Q14 CH8-LO4

Match each part of an HPLC to its job.

pump → pushes the mobile phase at high pressure; loop injector → delivers an exact volume of sample; C18 column → separates the components; UV detector → signals each compound as it leaves

Harvey 12.5.

Q15 CH8-LO4

Match each GC detector to what it is best for.

FID → almost all organic compounds; ECD → halogenated compounds; TCD → everything, but less sensitive; MS → identifying unknowns

Harvey 12.4.

Q16 CH8-LO4

Temperature programming in GC means:

A. the oven heats up during the run

B. the injector is cooled

C. the carrier gas is changed

D. the sample is frozen first

Low- and high-boiling compounds both give sharp peaks.

Pharmacy scenario · The slimming capsule

Q17 CH8-LO5

The quickest, cheapest first screen for a hidden drug in the capsule is:

A. TLC against reference standards of suspected drugs

B. GC of the capsule shell

C. ion-exchange chromatography

D. size-exclusion chromatography

Many samples and standards on one plate.

Q18 CH8-LO5

To confirm the identity of the hidden drug beyond doubt, the lab would use:

A. LC–MS or GC–MS

B. paper chromatography

C. a melting point only

D. iodine vapour

MS gives the mass and fragments of the molecule.

Q19 CH8-LO5

For the paracetamol tablets, which test shows whether the drug has broken down?

A. HPLC: less paracetamol and new breakdown peaks

B. GC residual solvents

C. a co-spot only

D. ion chromatography of chloride

A stability-indicating HPLC method.

Q20 CH8-LO5

Explain to Puan Rosnah, in plain words, how the lab can tell what is inside her capsule.

The lab dissolves the capsule contents and puts a tiny spot on a plate, next to spots of known drugs. A solvent rises up the plate and carries each substance a different distance. If a spot from her capsule travels exactly as far as a known slimming drug, that drug is probably inside. Then a more powerful machine (LC–MS) confirms it and measures how much there is.

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